

Rheology and early age evaluation of 3D printable cement-limestone filler pastes with nanoclays and methylcellulose

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ABSTRACT

Nanoclays (NC) are known to increase the structural build-up of cement-based materials (CBM) while methylcellulose (MC) is used to increase open time and workability, helping to prevent cold joints in 3D Printing applications. The combination and compatibility of both NC and MC could lead to the design of new 3D printable CBM able to tackle all the manufacturing process requirements. The aim of the study is to evaluate the effect of sepiolite nanoclay and MC on rheology and early age (EA) parameters of cement-limestone filler pastes for 3D printing applications, identifying fresh structural build up and setting mechanisms. The effect of two cement types with coarser and finer particle size distribution (CEM I 42.5-R and CEM I 52.5-R, respectively) was considered. Pastes rheology was characterized by Cone Penetration Test (CPT) and Dynamic Shear Rheometer (DSR). EA was monitored by P- and S-waves Ultrasonic Pulse Velocity (UPV), internal temperature, capillary pressure and drying shrinkage. Fresh Shear Yield Stress (τ_0) and shear elastic modulus (G), effective thixotropy (A_{thix}), critical strain (γ_{crit}) and their evolution over time were calculated from CPT and DSR experimental results. A limit of usability was defined for τ_0 up to 20 kPa for CPT with a 30° tip cone. Although pastes with both NC and MC showed thixotropy, NC reduced γ_{crit} while MC strongly increased it. NC significantly increased A_{thix} and ΔG with coarser particle size cement when compared to the finer cement. MC delayed hydration reaction and setting, while NC did not affect early age stages. On the other hand, combining NC and MC increased superplasticiser demand and reduced effective thixotropy (A_{thix}). Finally, an integrated overview of the physical and chemical mechanisms involved in rheology and early age properties of cement-based systems and the key experimental parameters for their identification was proposed.

1. Introduction

Digitalisation of processes is one of the great challenges towards sustainability in construction industry. Many additive manufacturing approaches have been developed in the last decades as for example extrusion techniques, binder jetting or particle bed [1]. Among digital fabrication technologies, extrusion-based 3D printing of cement based materials (3DPC) has stood as one of the most promising ones. Implementing 3DPC for large scale applications brings great benefits regarding to topological optimisation, reduction of resources consumption, safety in the construction site, cost reduction or expanding new horizons in architectural formal design [1–3]. However, 3DPC still faces issues related to material suitability for construction requirements. To be 3D printable, cement based materials (CBM) should be pumpable, extrudable and buildable by rapidly gaining strength and stiffness [3]. Buildability is related to material's structuration rate or thixotropy

index (A_{thix}) to prevent collapse [4,5], while the shear and compressive elastic modulus evolution over time would govern the dimensional stability [3,6].

CBM rheology is related to soft colloidal interactions, resulting from physical phenomena (i.e. Van der Waals forces and electrostatic attraction), and products nucleation as a result of chemical hydration of cement grains [7]. Although there are many types of superplasticizers used for CBM, e.g. naphthalene-based or lignin-based SP, the use of Polycarboxylate ether-based (PCE) SP is widely extended and allows to achieve a tailored rheology of CBM, greatly reducing water demand [8, 9]. Nevertheless, SP has an impact also on hydration kinetics and early age CBM parameters, delaying setting and hardening [10,11].

Another group of materials that are used to achieve the rheological requirements for 3DP are Rheology modifying admixtures (RMA) [12, 13]. Among RMA, nanoclays (NC) have proven to enhance CBM extrudability and buildability, developing high thixotropy and

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increasing shear strength and structural build up without significant changes in viscosity [14,15]. NC also contribute to shape stability due to a stiffening effect [14,16–20], the increase of static yield stress due to colloidal interactions [19] and the promotion of hydrates nucleation and flocculation [14,17,19], increasing hydration heat due to its seeding effect [16,21].

Attapulgite, a magnesium aluminium phyllosilicate - palygorskite, is the most investigated NC [19], although sepiolite (also a palygorskite but with larger needle-shaped particle size) has also been used for CBM rheology control [22–26]. However, some drawbacks have been described, as excessive stiffening during extrusion and filament tearing, increase of SP requirement or cohesiveness reduction, which might become a limitation, in terms of workability and filament continuity. [16,17].

These issues can be addressed with viscosity modifying admixtures (VMA) which are widely employed in 3DP applications to enhance CBM stability [3,12,27,28]. Cellulose ethers-based VMA are one of the most extensively used type [8,9,16,29]. However, VMA increase deformability, as they weaken van der Waals attractive forces network [27], and delay CSH nucleation due to cement adsorption [16,27,29] compromising 3DP buildability.

Combining NC and VMA has been proposed by other authors as a solution to achieve CBM rheological requirements for 3DP, each one compensating for the disadvantages of the other [16,23,24]. However, further research is needed on the compatibility, synergies and interactions of organic and inorganic admixtures.

The small particle size of RMA and VMA and their surface interactions with cement and supplementary cementitious materials (SCM) grains are behind their efficiency on CBM rheology [21]. Cement fineness increases thixotropy due to its higher specific surface area and is related to hydration heat at EA and setting [30–33]. Among SCM, limestone filler is widely used as a low cost and available product [34] which improves particle packing effect, reduces hydration heat, and improves workability and geometrical stability [35].

Although measuring the static shear yield stress (τ_0) and its evolution over time, related to A_{thix} , is essential for digital fabrication, recent studies highlighted the interest of also evaluating critical strain (γ_{crit}) as a useful parameter [7,36–39]. Some authors associated γ_{crit} to the critical deformation at flow onset [38,39] while other studies associate γ_{crit} to the limit of the linear viscoelastic region [36].

CBM rheology can be measured in the laboratory with a Dynamic Shear Rheometer (DSR), using either a Constant Shear Rate protocol (CR) [25,37,38,40] or a Controlled Shear Stress protocol (CS) [37,41]. Among other possibilities, Vane geometry has been demonstrated to perform well for 3DP mixtures [23,42].

Despite DSR is an accurate method to measure rheological parameters of CBM, fast and easy to run field-oriented tests that can be used for quality control are still limited. Cone penetration test (CPT) has been used in determining CBM fresh performance [23,37,43–46].

Further than rheology, other early age parameters related to cement hydration, material stiffening and dimensional changes have to be evaluated to understand the formation of the hardened CBM microstructure [47,48]. Material's internal temperature (T) allows monitoring the hydration process and provides information of the setting progress [49–53]. Ultrasound pulse velocity (UPV) can be used to follow the stiffening and setting [54–56]. Shrinkage (S_{hr}) and capillary pressure (P_{cap}) can also be used as control parameters of water kinetics dimensional stability during early age [50,57–61].

The aim of this study was to assess the effects of NC and MC on rheology and early age (up to 24 hours) of 3DP flowable pastes for layered extrusion systems. The effect of cement grain size was analysed, combining laboratory and field-oriented rheological tests with an EA test set-up.

Table 1

Cement nominal parameters.

	CEM 42.5	CEM 52.5
Clinker (%)	95	95
Limestone (%)	5	5
SO ₃	3.3	3.4
Chlorides	0.01	0.01
Blaine fineness (cm ² /g)	3600	4800

2. Materials and methods

2.1. Materials and mix design

Six different paste compositions were designed using two types of cement, CEM I 42.5 R and CEM I 52.5 R named according to the European Standard, blended with limestone filler Betocarb® P1-DA (85 ± 5 % under 63 µ), supplied by Omya Clariana SL. Cements were manufactured at the same plant and their properties are summarised in Table 1, where the different grain size and Blaine fineness can be observed.

Sepiolite powder (SEP), a palygorskite nanoclay with needle shaped particles and a BET surface of 316 m²/g, in 1 and 2 % by cement weight was used as RMA. On the other hand, 0.25 and 0.125 % by cement weight of methylcellulose (MC) was used as VMA. Paste compositions are summarised in Table 2.

Mixtures compositions were designed based on paste consistency measured by cone penetration test (CPT) with a penetration depth of 18 mm corresponding to an estimated shear yield stress (τ_0) of 1.000 Pa. A Polycarboxylate Ether-based superplasticizer (SP) presented in powder form was adjusted to achieve the target initial yield stress while water to cement ratio (w/c) remained constant at 0.36 for all the pastes.

Mixing began with all the dry components in a planetary mortar mixer for one minute, water was added afterwards and mixing went on for 5 more minutes.

2.2. Rheological characterization

Cone penetration test (CPT) is a field-oriented test which measure the penetration depth of a 30° conical tip, weighing 80 g, dropped on a paste sample and relates it to the paste rheological properties (Fig. 1). CPT was carried out five times on paste samples left at rest and stirred before tested at 10, 20, 30, 45, 60, 75, 90, 105 and 120 minutes to determine reversible and non-reversible time-dependent structural build up, respectively [4,25]. Test protocols and parameters are summarised in Fig. 1a. Shear yield stress (τ_{0-CPT}) was calculated according to (Eq. 1) [45]

$$\tau_{0-CPT} = \frac{m \cdot g \cdot \cos\theta}{\pi \cdot h^2 \cdot \tan\theta} \quad (1)$$

where m is the cone mass, g is the acceleration of gravity, θ is the conical angle, and h is the penetration depth (mm).

A Searle-type HAAKE MARS II® Dynamic Shear Rheometers (DSR) was used to determine τ_{0-DSR} more accurately on paste samples poured in a ribbed cylindrical cup and tested every 10 minutes with a vane geometry, as showed in Fig. 1b. A geometry factor (A) was calculated according to (Eq. 2). [62]

$$A = \frac{1}{4\pi R_i^3 \left(\frac{L}{2R_i} - \frac{1}{3} \right)} \quad (2)$$

Two DSR rotational test protocols were conducted to determine yield stress: a controlled stress protocol (CS) that increases shear stress in a constant ramp from 100 Pa to 10.000 Pa for 300 s, and a controlled rate protocol (CR) with a predefined constant shear rate at 0.05 s⁻¹ for 200 s (Fig. 1b). Some samples were stirred before testing at all testing times (named CR-S and CS-S) while other were left at rest until tested (named

Table 2

Paste compositions (in g) and initial rheological parameters.

	Composition	CEM I 42.5 R	CEM I 52.5R	Limestone Filler (g)	w/c	HRWRA (%wt)	SEP (g)	MC (g)	τ_{0-CPT} (kPa)	τ_{0-DSR} (kPa)
CEM 42.5	REF	1404	-	468	0.36	0.05 %	-	-	0.56	0.75
	SEP1	1404	-	468	0.36	0.30 %	14	-	0.87	2.54
	SEP2	1404	-	468	0.36	0.60 %	28	-	1.17	3.29
	MC1	1404	-	468	0.36	0.125 %	-	1.8	2.7	2.09
	MC2	1404	-	468	0.36	0.25 %	-	3.5	1.17	2.84
	SEP+MC	1404	-	468	0.36	0.90 %	28	3.5	1.29	2.84
CEM 52.5	REF	-	1404	468	0.36	0.20 %	-	-	0.66	0.99
	SEP1	-	1404	468	0.36	0.50 %	14	-	0.69	1.5
	SEP2	-	1404	468	0.36	0.80 %	28	-	0.85	1.79
	MC1	-	1404	468	0.36	0.30 %	-	1.8	1.03	3.29
	MC2	-	1404	468	0.36	0.45 %	-	3.5	0.53	2.99
	SEP+MC	-	1404	468	0.36	1.50 %	28	3.5	0.75	1.79

CR-R). In both protocols, an initial elastic phase of the stress-strain curve was obtained, where elastic shear modulus (G) was calculated [17].

Roussel's thixotropic model (Eq. 3) [4] is used to evaluate rheological parameters such as yield stress or thixotropy:

$$\tau(t) = \tau_0 + A_{thix1} \cdot t \quad (3)$$

2.3. Early Age monitoring

Early Age (EA) monitoring included several non-destructive experimental techniques (NDT) applied simultaneously on CMB samples of the same batch. Internal temperature (T), capillary pressure (P_{cap}), free shrinkage (S_{hr}), weight loss and transmitted P and S waves Ultrasonic Pulse Velocity (UPV) were simultaneously recorded on paste samples for 20 hours. A graphical overview of the experimental program is presented in Fig. 1c.

A modular multi-test casting system was designed and patented as a method for monitoring EA materials properties (Patent application number: P202331019), which consists of a modular mold that adapts to any sample size and testing requirement (Fig. 1c). Four screwed steel rods assemble the central U shape body that contains the material sample and a couple of headwalls adapted to the specific sensor or transducer type, as can be seen in Fig. 2.

Internal temperature (T) was monitored using two type K thermocouple immersion probes, exposed to laboratory conditions (20 ± 2 °C and 60 ± 10 % RH) [3].

UPV was monitored with 54 kHz p-wave transducers connected to a PUNDIT LAB+ ® device on 80 mm samples and 250 kHz p- and s-waves transducers connected to a CNS Farnell PUNDIT PLUS® device on 40 mm samples, every 10 minutes.

Capillary pressure (P_{cap}) was monitored with four pressure sensors located at two different depths as described in [51,57].

Free shrinkage (S_{hr}) was tested with a sample cast in a 250 mm modulated mold, with one side fixed to the mold and the other set free to displace and connected to a LVDT Mitutoyo ® 575–121 (± 0.02 mm accuracy). A 3 m/s air flow was applied on the upper sample surface for the first 6 hours to force water evaporation. The mass loss of the sample was continuously recorded by a balance placed under the shrinkage apparatus.

To perform a comparative study of the admixture's effects on early ages, all the EA variables were processed using the first derivative to identify EA patterns and stages [32] [49–51,53–55].

3. Experimental results

3.1. Rheological assessment: CPT and DSR results

The evolution of τ_{0-CPT} up to 120 minutes is plotted in Fig. 3. The use of a CEM I 52.5-R cement type produced a higher growth of τ_{0-CPT} value compared to CEM I 42.5-R cement pastes, except for the mixture with

2 % of sepiolite (SEP2). All admixtures increased the final τ_{0-CPT} after two hours compared to REF, especially for mixtures MC1, regardless of the cement fineness. On the other hand, the combination of MC and NC resulted in a very poor structuration rate.

Fig. 4 presents the DSR yield stress (τ_{0-DSR}) and elastic modulus (G) evolution up to 60 minutes. As expected, coarser cement (CEM I 42.5-R) showed smaller τ_{0-DSR} and G gains than finer cement (CEM I 52.5-R) due to the smaller surface area. However, an exception is observed in presence of sepiolite, when stiffening and structuration rates are clearly higher in 42.5 systems than 52.5, especially for G .

Fig. 5 compares the rheological parameters of samples stirred before tested and samples left at rest until tested. Generally speaking, pastes at rest showed larger values of τ_{0-DSR} and G .

NC and MC enhanced fresh structuration rate, but the thixotropic effect of MC was more efficient in lower amounts, as shown in Fig. 4 and Fig. 5. On the other hand, combining NC and MC did not significantly improve the rheological performance compared to REF and no synergies were founded for the studied proportions.

The evolution of G indicates that sepiolite produced a remarkable stiffening effect, especially within 42.5 R cement pastes, rather than on yield stress. By contrast, higher addition of MC (MC2) reduced G evolution over time, compared to MC1.

3.2. Early age monitoring

Fig. 6 plots the EA parameters evolution up to 20 hours and the analysis methodology to define EA key points in each case, which are presented in Table 5.

During initial minutes, pastes consisted of a colloidal suspension in which ultrasonic pulses (Fig. 6a) propagate through the aqueous phase, remaining constant around 1000 m/s. The formation of a weak solid microstructure produced changes defined by three turning points in velocity growth trends (B-D) [54,55]. After point D, a final stabilisation period begins, reaching a final UPV around 3300–3500 m/s.

The inclusion of MC and NC showed a delay in UPV curves when compared to REF, regardless cement type. SEP2 presented the smallest lag, while MC clearly induced delay proportionally to its concentration. The combination of NC and MC caused an even larger delay when added in 52.5 systems but moved them forward in 42.5 pastes. Finer cement (52.5) showed a larger delay than coarser cement (42.5). However, REF mixture showed larger final UPV than pastes with NC and MC, especially with finer cement particles. Pastes with 42.5 cement reached a final velocity around 3500 m/s while 52.5 pastes rose to over 3600 m/s.

Five typical stages can be drawn at the early age temperature curve, defined by 4 key points (Fig. 6b) [32,49,53]. First stage starts with a temperature peak related to alite hydration [49]. After a fast deceleration known as dormant period, a minimum point is reached (A) which defines the initial setting and the beginning of the acceleration period [32]. Afterwards, two maximum points appear in the first derivative curve: (B) reached after a smooth increase in temperature at the

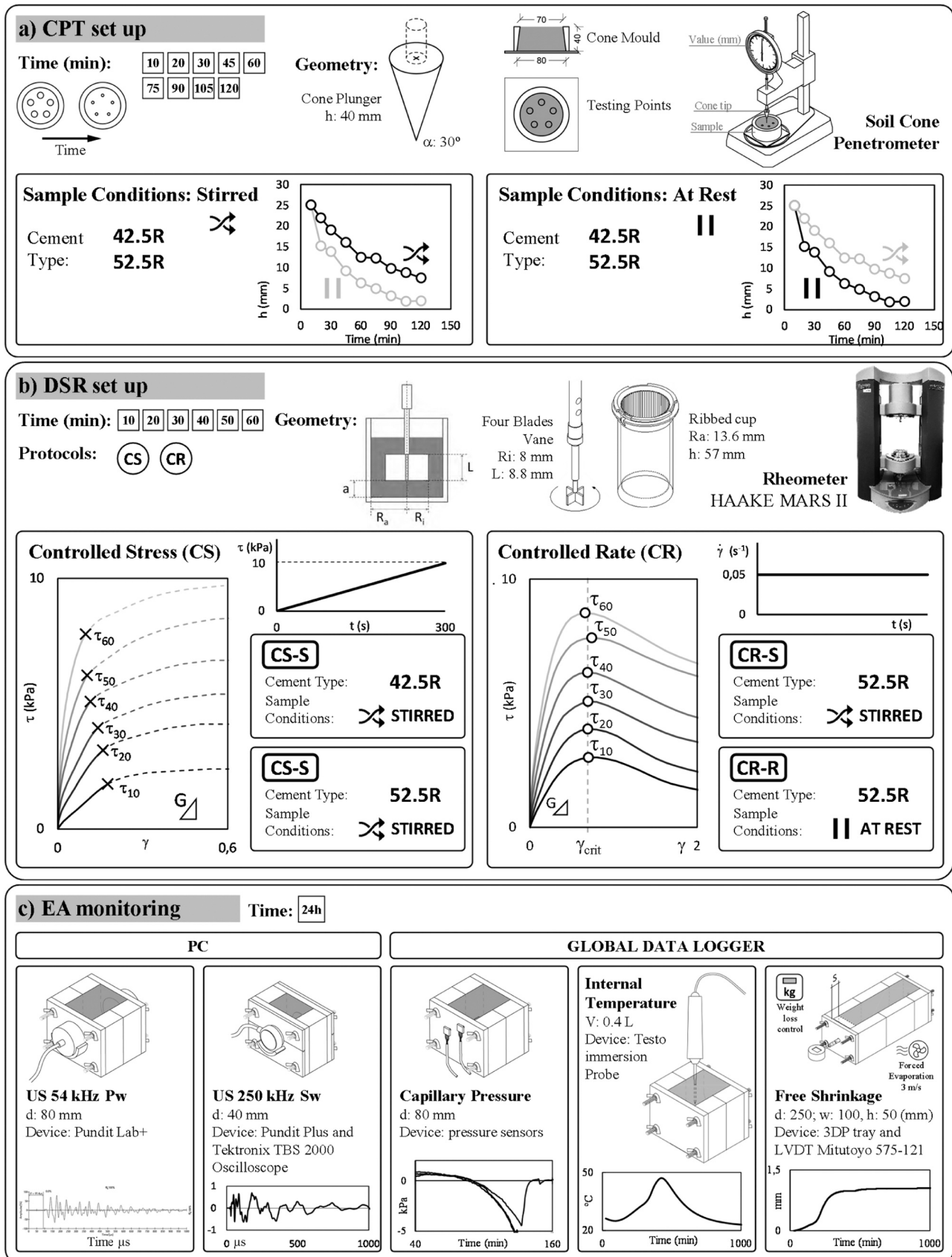


Fig. 1. Experimental program: a) Cone penetration test (CPT); b) DSR protocols and conditions; c) Early age testing setup, using the multi-test modular casting system (patented: Patent application number: P202331019).

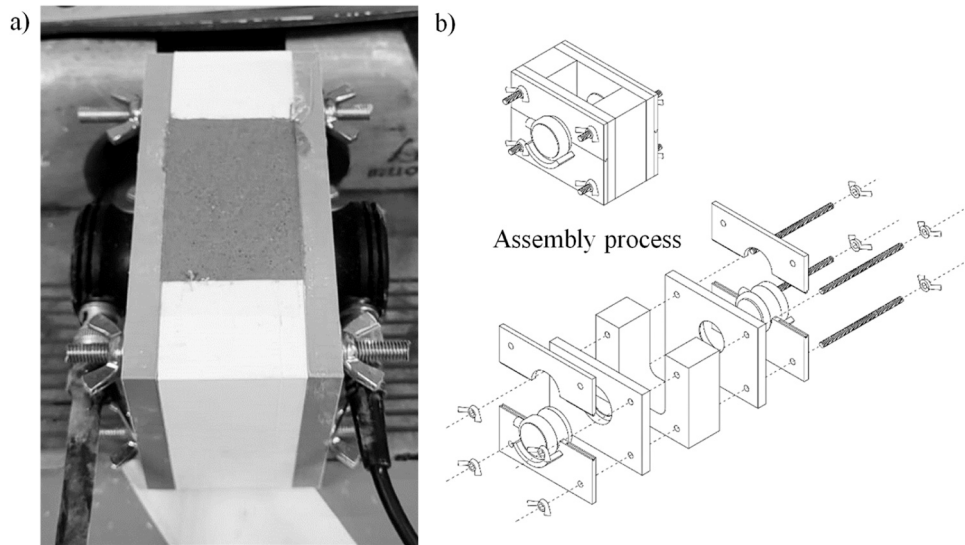


Fig. 2. UPV monitoring setup of cement pastes: a) paste sample poured in the multi-test casting system (Patent application number: P202331019); b) UPV testing assembly of the modular multi-test casting system (patented).

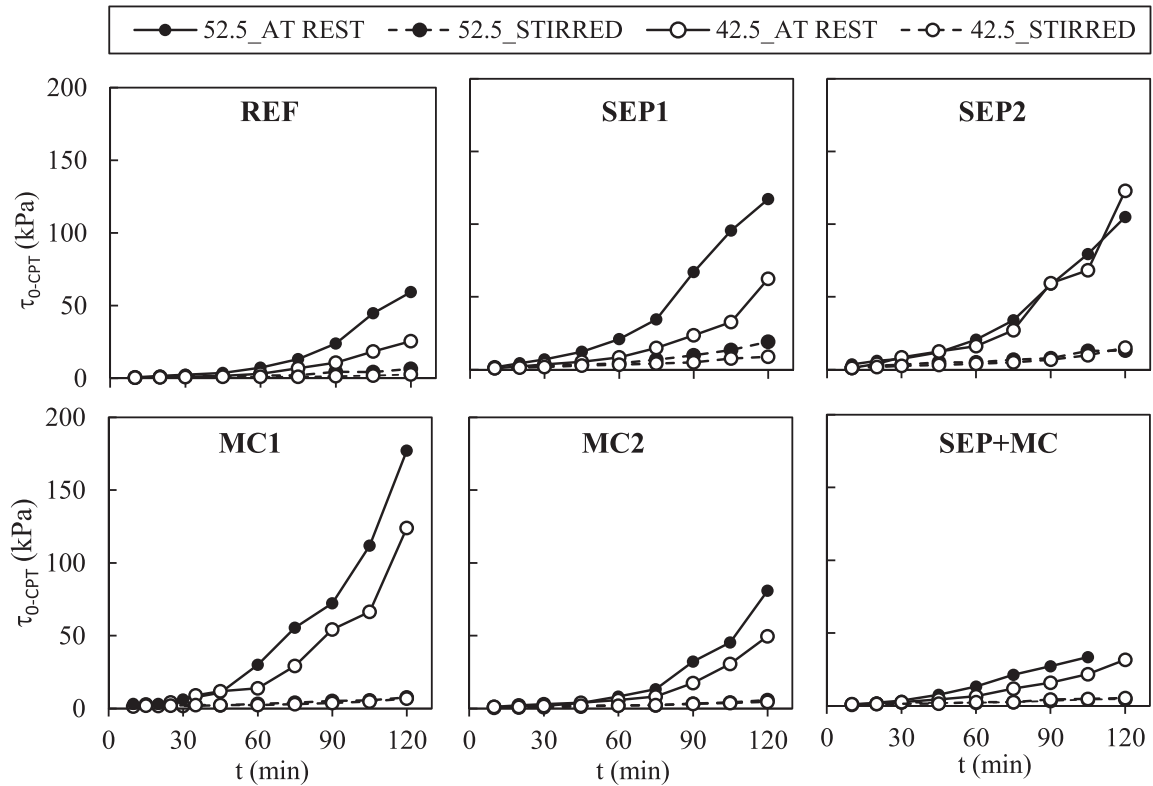


Fig. 3. Shear yield stress evolution calculated from CPT (τ_{0-CPT}) up to 120 minutes.

beginning of the acceleration period and (C) after an abrupt increase in heat release. This point marks the beginning of a deceleration period until the minimum heating rate (D) that leads to temperature stabilisation.

Sepiolite did not influence EA temperature timing regardless cement fineness, while MC delayed heat release significantly. Mixture combining SEP and MC enlarged delay on 52.5 pastes but shortened processes on 42.5.

Finer cement delayed thermal evolution and reduced peak temperatura in comparison to 42.5 samples, probably due to the larger amount

of SP. REF showed the higher peak (47°C) in 42.5 samples, nearly followed by SEP2 (43°C). MC1 and MC2 samples showed similar maximum temperatures (around 35°C). However, REF showed the higher reduction of peak temperature on 52.5 (30°C), being even lower than SEP2 (37°C) with this type of cement and similar to samples with MC.

Shrinkage produced two types of curves: S-bend curve (mixtures REF and SEP2) with an initial accelerated increase, reaching an inflection point and a last decreased growth, reaching a tableau [50,51] and C-bend curve, showing only the last S-curve stage and bringing forward shrinkage beginning (mixtures with MC). Besides, the effects of the

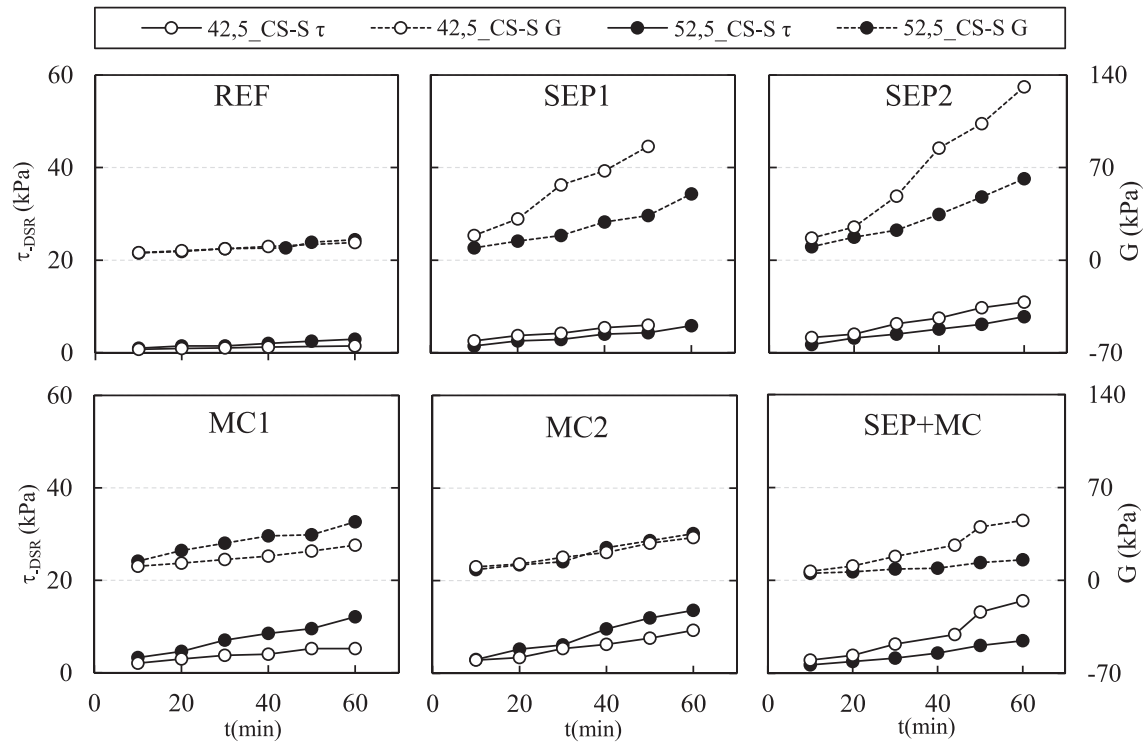


Fig. 4. Influence of cement type on DSR shear yield stress (τ_{DSR}) and shear modulus (G) evolution on samples stirred before tested, up to 60 minutes.

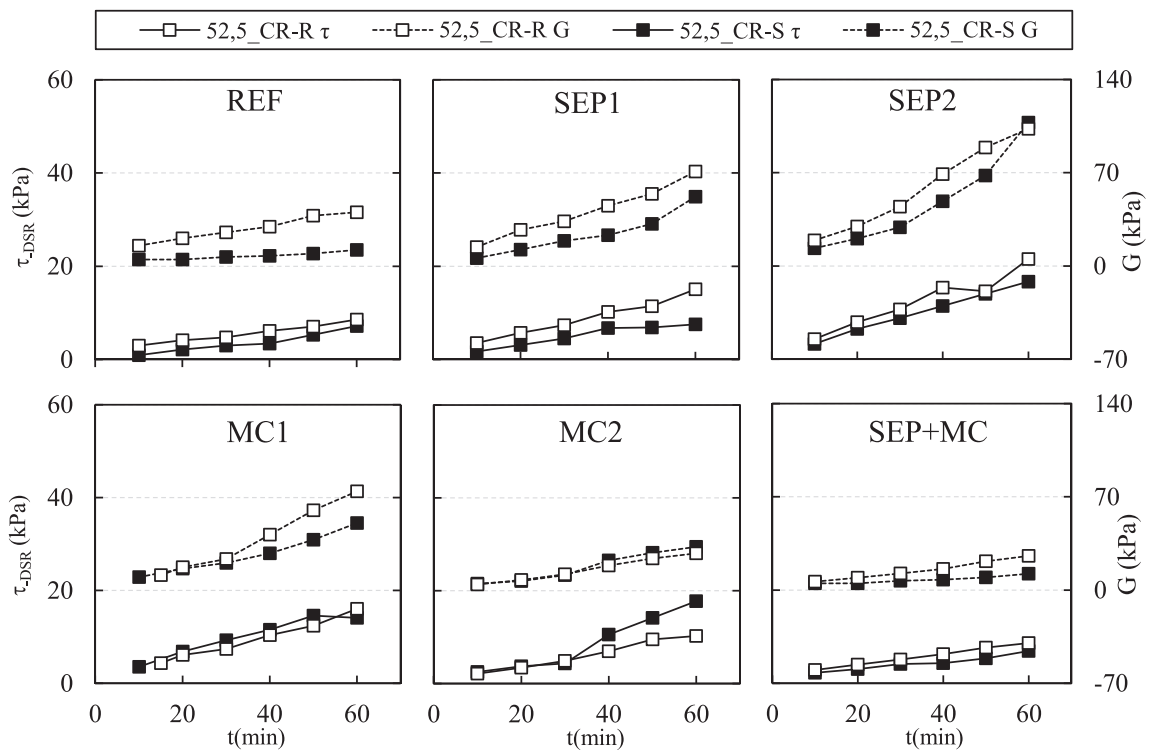


Fig. 5. DSR shear yield stress (τ_{DSR}) and shear modulus (G) evolution of samples stirred before tested and left at rest, up to 60 minutes.

admixtures on shrinkage were not influenced by cement fineness. REF presented higher shrinkage in 52.5 systems [30] and SEP2 significantly increased the total shrinkage up to 4 mm/m. On the other hand, final shrinkage was lower in MC pastes and SEP+MC move the shrinkage

onset forward compared to REF.

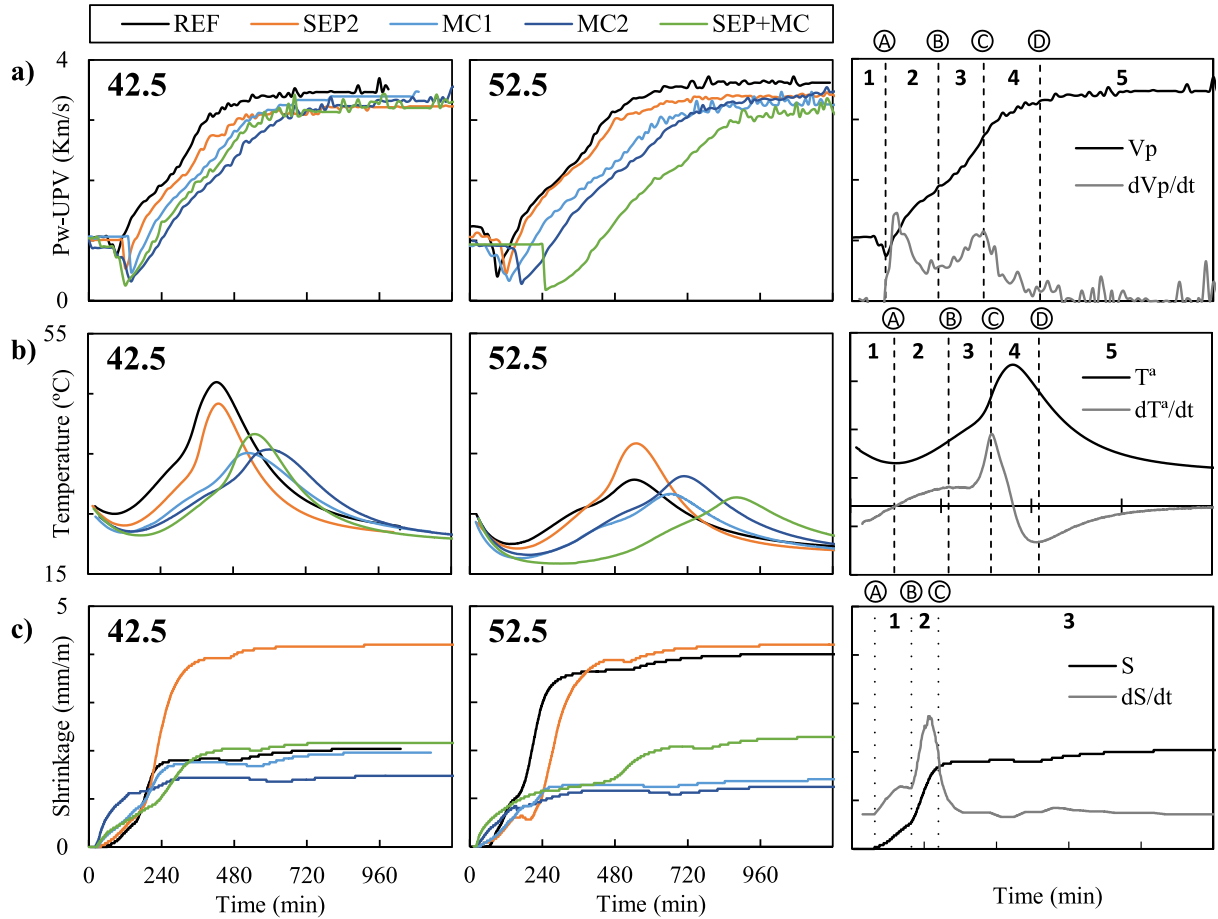


Fig. 6. Monitoring of early age parameters, ultrasonic pulse velocity (UPV), temperature (T) and free shrinkage (S), and analysis criteria for EA stages definition.

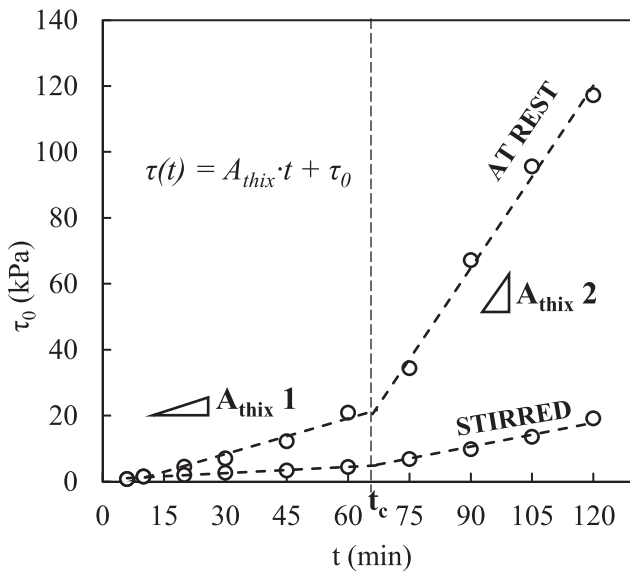


Fig. 7. Bilinear thixotropic modelling of τ_0 calculated from CPT (τ_{0-CPT}).

4. Analysis and discussion

4.1. Influence of NC and MC on rheology and buildability

Fig. 7 plots the bilinear model of A_{thix} on samples at rest and stirred,

corresponding to static yield stress estimated from penetration values and adjusted according to Roussel's model [4]. Initial τ_{0-CPT} grew at a certain rate referred to as A_{thix1} until a characteristic time t_c , when a second stage characterized by a higher structuration rate A_{thix2} [5]. The value of t_c is determined by the mathematical intersection of both straight lines defined in Eq. (4):

$$\tau(t) = \begin{cases} \tau_0 + A_{thix1} \cdot t, & t \leq t_c \\ \tau_{tc} + A_{thix2} \cdot (t - t_c), & t > t_c \end{cases} \quad (4)$$

This bilinear trend is suitable to both at rest and stirred material situations and was considered for all pastes. Stirred structuration rate can be related to irreversible thixotropy [3,23]. Table 3 presents both thixotropy indexes from bilinear model on both conditions, samples at rest and stirred, ($A_{thix-R1}$; $A_{thix-R2}$; $A_{thix-S1}$; $A_{thix-S2}$; respectively), also t_c for each mixture. Moreover, two new coefficients to explain the difference between at rest and stirred structuration rates on both linear adjustments were included in Table 3 ($C_{thix1} = A_{thix-R1}/A_{thix-S1}$ and $C_{thix2} = A_{thix-R2}/A_{thix-S2}$). C_{thix1} and C_{thix2} can be related to the degree of reversibility of the mechanisms responsible of the stiffening, and hence they also point out if the structural build up is caused by physical or chemical processes [5,7,25]. Higher C_{thix1} and C_{thix2} can be associated to material workability and open time window.

MC1 presented a significantly different behaviour, showing a high $A_{thix-R1}$ and C_{thix1} ratio. After 2 hours of resting time, MC1 reached a τ_{120} of 177 and 124 kPa on 52.5 and 42.5 samples, respectively, while remaining around 7 kPa on stirred MC1 sample. Pastes including MC present higher C_{thix1} and C_{thix2} values compared to the reference, meaning that MC admixture enhances workability, but not necessarily

Table 3

Effective thixotropy values (A_{thix}) calculated from CPT results.

	SAMPLES	$A_{thix\ R1}$ (Pa/min)	$A_{thix\ R2}$ (Pa/min)	$A_{thix\ S1}$ (Pa/min)	$A_{thix\ S2}$ (Pa/min)	t_c (min)	C_{thix-1}	C_{thix-2}
CEM 42.5	REF	49	424	8	37	75	5.8	11.6
	SEP1	148	838	51	113	73	2.9	7.4
	SEP2	306	1979	58	221	76	5.3	8.9
	MC1	267	1717	8	82	69	32.7	20.9
	MC2	105	915	20	46	75	5.3	20.0
	SEP+MC	110	399	40	40	62	2.7	-
CEM 52.5	REF	128	1062	21	88	73	6.0	12.1
	SEP1	458	1846	76	273	72	6.0	6.8
	SEP2	326	1430	82	161	62	4.0	8.9
	MC1	232	2083	29	68	51	8.0	30.8
	MC2	193	1738	29	79	75	6.8	22.1
	SEP+MC	359	359	32	32	-	11.1	-

Table 4

Effective thixotropy (A_{thix}) and Stiffening rate (ΔG) of pastes calculated from DSR tests.

		CEM 42.5		CEM 52.5	
		CS-S	CS-S	CR-S	CR-R
A_{thix} (kPa/min)	REF	0.01	0.04	0.12	0.11
	SEP1	0.09	0.08	0.12	0.22
	SEP2	0.16	0.12	0.26	0.32
	MC1	0.06	0.17	0.22	0.25
	MC2	0.13	0.22	0.33	0.17
	SEP+MC	0.26	0.10	0.09	0.12
ΔG (kPa/min)	REF	0.16	0.31	0.65	0.52
	SEP1	1.71	0.78	0.84	1.07
	SEP2	2.41	0.93	1.81	1.77
	MC1	0.31	0.54	0.79	1.43
	MC2	0.46	0.57	0.61	0.49
	SEP+MC	0.80	0.20	0.14	0.38

buildability [16,28]. This effect can be explained by the fact that the material had very low $A_{thix-S1}$ and $A_{thix-S2}$, which implies that the mechanisms responsible for this initial structural build up are mainly physical in nature (reversible thixotropy) [7]. The adsorbed polymeric particles prevented hydrates nucleation during the initial time. Nevertheless, the colloidal interactions responsible of the structural build up increased the resistance to penetration over time. Brumaud et al. suggested that cellulose ethers can collapse van der Waals forces and create a new interaction network based on the bridging effect between cement grains by the adsorbed particles [27]. This physical interaction could explain the high τ_0 increase of MC1 at rest condition but its low growing on stirred condition (Fig. 3 and Table 3).

On the other hand, SEP2 rose τ_0 to 105 and 123 kPa after 2 hours of resting time while stirred sample barely reached 15 kPa, doubling irreversible thixotropy. This could be related to a larger production of early hydrates promoted by the filling and seeding effect [17,21]. NC also reduced structuration rates C_{thix1} and C_{thix2} , reducing workability and the open time window.

In the proposed compositions, pastes combining NC and MC did not reach t_c during the test period, meaning that this composition remained as a soft-plastic material within the initial 120 minutes. This is explained by the high concentration of polymeric additions (large amounts of MC and SP) which retarded the nucleation of hydration products. For this reason, SEP+MC did not have a C_{thix-2} index, and C_{thix-1} remained low. Considering the application for extruded layers deposition, this material does not provide an adequate structural build-up for 3DP. Nevertheless, further research on the optimum mix proportion combining NC and MC could arise promising results, known the effects produced by these two components in lower amounts and the influence of cement fineness. The loss of workability caused by a lower SP efficiency could be related to the reduction of cement volume fraction due to the high concentration of admixtures, in addition to the interactions between NC and MC.

Regarding cement type, 52.5 showed higher thixotropy than 42.5

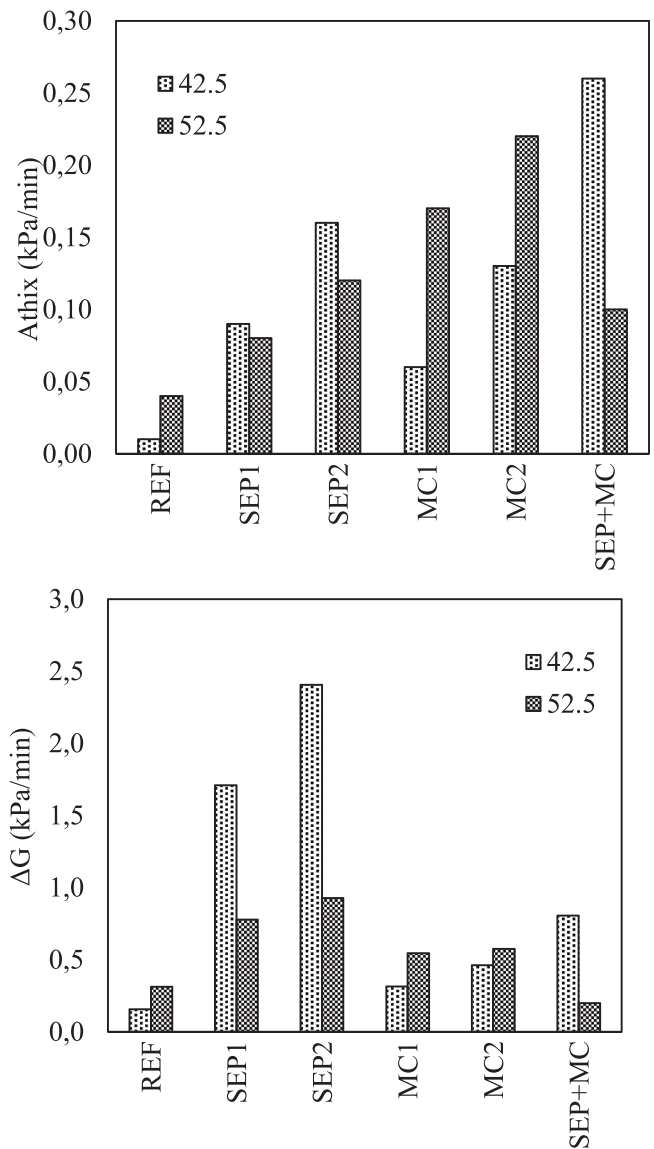


Fig. 8. Effective thixotropy (A_{thix}) and Stiffening rate (ΔG) from DSR tests.

samples owing to its smaller particle size and therefore a higher Blaine fineness (Table 1), increasing surface interactions and nucleation sites [31,32]. Nevertheless, SEP2 presented an exception in 42.5 pastes, showing even larger thixotropy than in 52.5 systems. This suggests that the thixotropic effect of NC is boosted with larger cement particle size.

Table 4 presents growth rates of shear stress (τ) and shear modulus

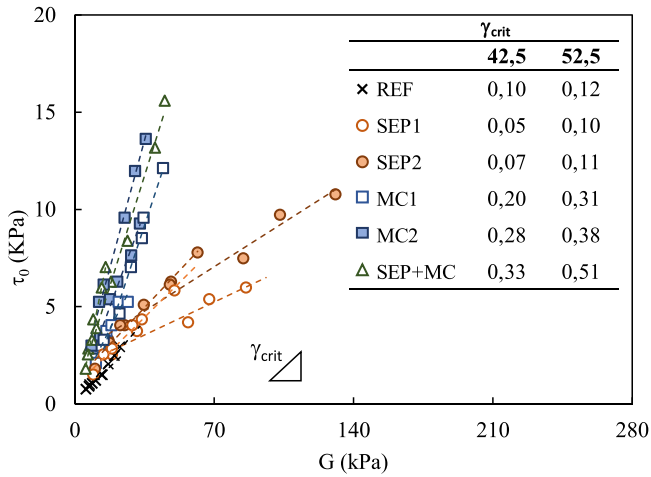


Fig. 9. Critical shear strain (γ_{crit}) of fresh cement pastes.

(G) over time, evaluated with CS and CR DSR protocols. Both NC and MC increased thixotropy of pastes (A_{thix}) according to their proportioning, but they did not have the same effect on ΔG , as plotted in Fig. 8. NC increased more stiffness than strength, especially in 42.5 cement samples. Mixtures with sepiolite showed higher A_{thix} and ΔG in 42.5 cement samples than in 52.5, related to the particle size distribution of binder systems [16,19,21]. Despite the higher final strength of a 52.5 cement type, the large requirement of SP is an important aspect to consider for 3D printing applications. As shown in this study, many rheological and early age parameters are very sensitive to changes in SP dosage. Consequently, the long-term benefits of using finer cements may not compensate the loss of workability or geometrical stability, considering that these materials could be used not only for structures, but also for any other architectural application. Critical strain (γ_{crit}) defines the maximum deformation at flow onset, when the static τ_0 is reached, and relates τ_0 and G, according to (Eq. 4):

$$\tau_0 = G \cdot \gamma_{crit} \quad (4)$$

On mixtures evaluated, γ_{crit} is a specific attribute of each mixture and does not undergo substantial changes over the first 60 minutes. Therefore, assessing γ_{crit} was an effective way to characterise 3D printable pastes fresh mechanical performance and the effects produced by components, regardless of strength and stiffness [7,36,48].

As mentioned above, sepiolite addition increased G more significantly than τ_0 , while MC behaved on the opposite way. This means that pastes with NC presented lower γ_{crit} than the reference, and MC increased this parameter, as can be seen in Fig. 9 [16,27]. This aspect is

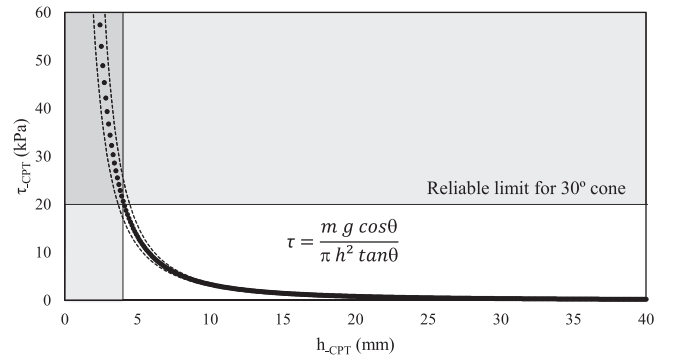


Fig. 11. Proposed limit of reliability for CPT with a 30° plunger.

of great significance for buildability, since higher γ_{crit} means larger deformations under the same load, and thus poorer dimensional stability and shape control [3,7,48].

NC and MC combined increased γ_{crit} , which could be related to the large SP amount necessary. Nevertheless, the combination boosted NC thixotropic effect with 42.5 pastes.

This fact is of great relevance in terms of optimal mix design combining both NC and MC. The blend of lower amounts of these constituents within 42.5 cement systems could yield interesting results, overcoming some of the mentioned drawbacks and achieving synergetic effects [16].

4.1.1. Limits of use of CPT as a field-oriented printability test

The formulation model used to obtain static τ_0 from CPT tests presented some issues concerning the estimation of material consistency for the pastes under study [45]. Some studies suggest that penetration resistance does not depend just on shear strength (τ) but also on compressive strength (σ) [23]. Viscosity could also play a role since falling time has an impact on the final depth value due to deferred penetration.

Fig. 10 presents a phenomenological hyperbolic adjustment between τ values measured in DSR and CPT penetration depth. It can be observed that the results did not fit perfectly with the model and showed variations depending on the paste compositions. In general, the theoretical model underestimates τ for penetrations over 7–8 mm and overestimates its value under this limit. Considering the model, a penetration depth under 4 mm would produce an error for τ higher than 5 % for each 0.1 mm which would increase exponentially for lower values. Accordingly, the authors propose a reliability limit of 4 mm of penetration or τ of 20 kPa for a 30° cone tip, considering lower penetrations out of the test range, as depicted in Fig. 11. To overcome this limit, a smaller cone

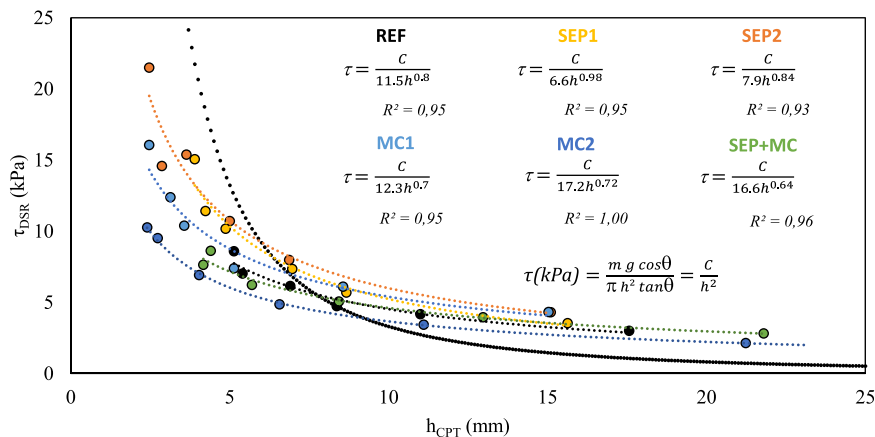


Fig. 10. Phenomenological modelling of τ_{DSR} values related to cone penetration depth in CPT (h) and theoretical model. [44].

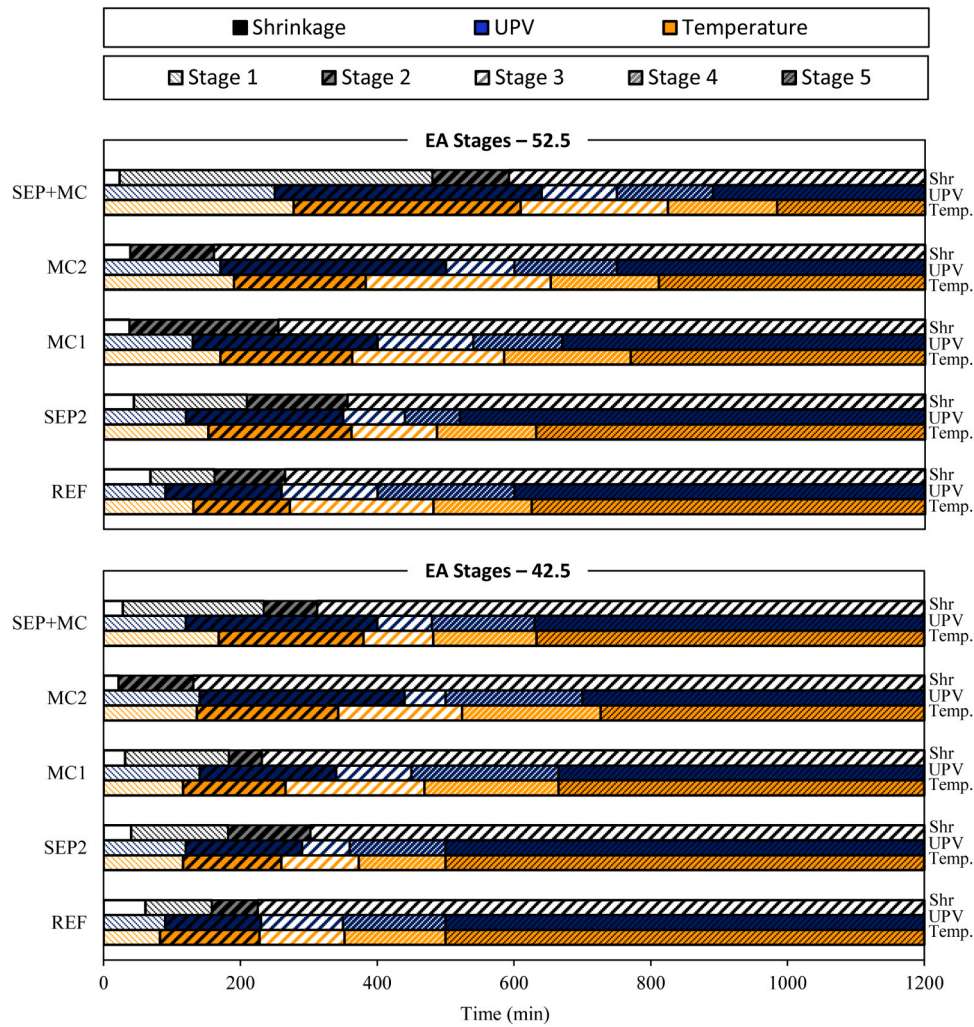


Fig. 12. Early age parameters evolution over time.

tip angle could be used, i. e. the reliability frame could be enlarged up to 55 kPa reducing the angle to 15°.

4.2. Effects of NC and MC on cement hydration

Fig. 12 compares the early age parameters (Shrinkage, UPV and Temperature) and stages defined for the different mixtures [52]. RMA and cement fineness produced changes in EA development. In 52.5 systems, UPV stages take place before than temperature stages. Hence, the hydration degree and the development of a solid microstructure depends on the type of cement and mixture composition.

MC have a clear influence on cement hydration, extending all the periods and delaying setting. This effect could help increasing the open time and avoiding cold joints, but also undermines buildability by delaying strength gain [16,27,49,53,54]. Pastes with NC also presented a small delay in the initial stages, which can be attributed to the higher concentration of SP [11,25].

NC and MC combined required larger amount of SP delaying hydration, especially on 52.5 cement (Fig. 12) [8,10,27]. As 42.5 systems required less SP, NC was more effective on enhancing hydration.

Against expectations, finer cement pastes showed lower hydration heat than coarser cement, delaying mechanical capacity [30–32]. It can

Table 5

Early age key points of cement pastes (min).

		Temperature				UPV				Shrinkage		
		A	B	C	D	A	B	C	D	A	B	C
CEM 42.5	REF	82.5	228	352	500	90	230	350	500	61	158	225
	SEP2	116	260	373	500	120	290	360	500	40	182	302
	MC1	116	266	469	665	140	340	450	665	31	183	231
	MC2	136	343	524	727	140	440	500	700	22	-	131
	SEP+MC	168	380	482	633	120	400	480	630	28	234	312
CEM 52.5	REF	130.5	272	482	625	90	260	400	600	68	162	265
	SEP2	152.5	362	487	632	120	350	440	520	44	209	356
	MC1	170	363	585	770	130	400	540	670	38	-	255
	MC2	190	383	653	811	170	500	600	750	39	-	161
	SEP+MC	277.5	609	824	984	250	640	750	890	23	480	592

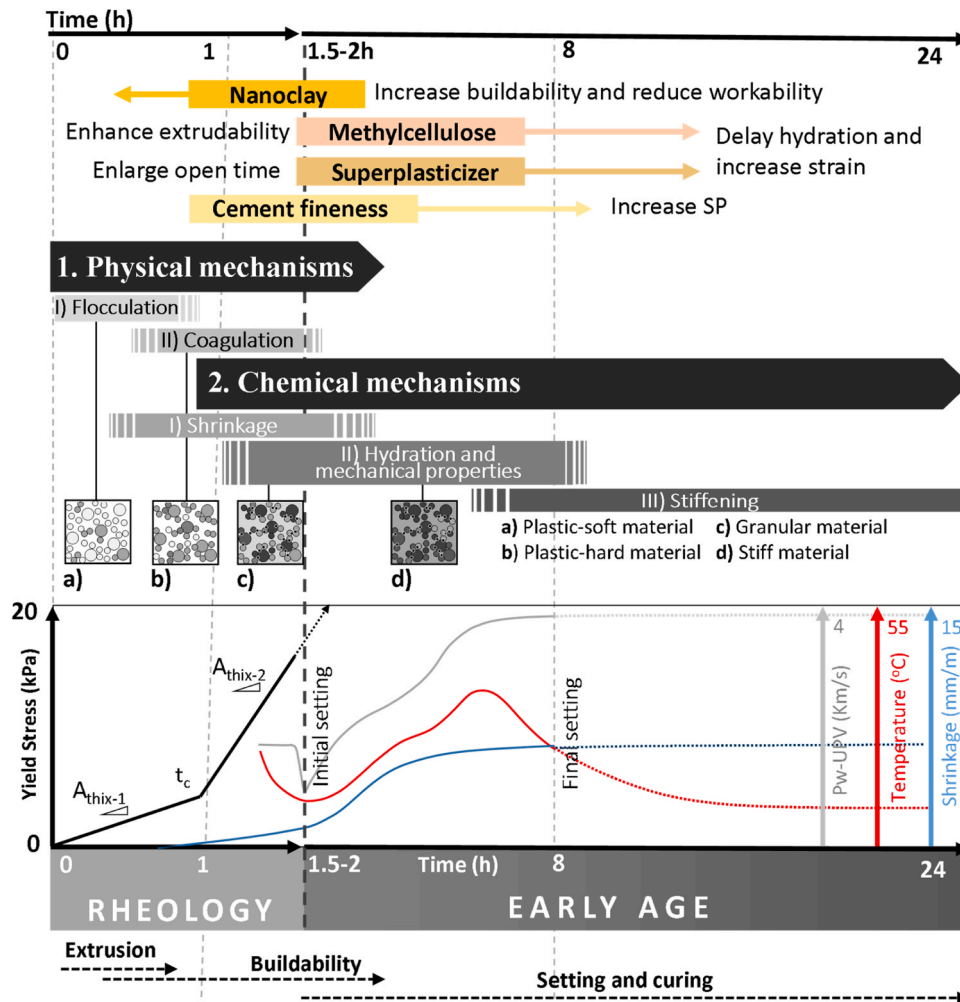


Fig. 13. Overview of physical and chemical mechanisms behind cement based systems changes at fresh state and early age: effect of paste components and rheology and early age parameters.

be explained by the larger amount of SP which has delayed setting and hydration kinetics [10,54]. Therefore, cement fineness is an important factor to consider in the design of 3DP composites, since it can significantly increase SP requirement delaying EA development.

In mixtures without MC, time of final shrinkage matches with the onset for UPV (stiffening) and temperature acceleration (cement hydration). Nevertheless, the addition of MC moved this period forward, earlier initiating shrinkage but also delaying final shrinkage, decreasing cracking risk.

Table 5 presents the key-points time for the three EA parameters. As final shrinkage in MC2 samples was reached before initial setting measured by temperature, it can be concluded that the underlying mechanisms of EA shrinkage are physical and related to those behind the reversible thixotropy of MC mixtures observed in CPT and DSR.

Reversible and irreversible thixotropy can be quantified by the difference between samples left at-rest and stirred before tested (Table 3). The delay produced by MC on hydration and mechanical properties can explain the low irreversible thixotropy and the high C_{thix} ratios. However, reversible thixotropy was mainly governed by flocculation and coagulation (colloidal interactions), rather than cement hydration, as it took place before initial setting.

Fig. 13 presents a graphical scheme that relates the physical and chemical mechanisms involved in fresh state rheology and early age parameters and the effect of the paste components. Colloidal interactions are involved in the physical mechanism (flocculation and coagulation) of material during the initial 2 h. On the other hand,

hydration starts after one hour and goes until 8 h due to chemical mechanisms. A scheme of cement paste evolution is presented from plastic-soft material to become in a stiff material.

5. Conclusions

This study presents the influence of sepiolite nanoclay (NC), and methylcellulose (MC), on 3D printing cement-limestone filler pastes rheology and early age parameters. Two different cement types (42.5 R and 52.5 R) were used to assess the effects of cement fineness with NC and MC. A superplasticizer (SP) was used to obtain the initial 3D printable target consistency. A Dynamic shear rheometer (DSR) and a cone-penetration test (CPT) were used to measure rheological parameters as shear yield stress (τ_0), structuration rate/effective thixotropy (A_{thix}), elastic modulus (G) and critical strain (γ_{crit}), and their influence on buildability. These laboratory test and field-oriented test were compared to assess its limits of use for pastes. On the other hand, ultrasonic pulse velocity (UPV), internal temperature, capillary pressure and shrinkage were monitored at early age (EA) to evaluate the influence of components on hydration kinetics and mechanical properties development. The following conclusions can be drawn:

- CPT is a useful field-oriented test method to assess the extrudability of cementitious materials. However, accurate models are still needed to determine the influence of compressive and shear strength in

resistance to penetration. A reliable limit was established in 20 kPa for a 30° tip cone angle.

- A reversibility coefficient responsible for stiffening of material (C_{thix1} and C_{thix2}) was defined comparing rheological parameters of samples left at-rest and stirred before tested.
- MC showed a high potential for enlarging open time, since pastes with MC presented a high thixotropy at rest with a very low structuration rate on stirred conditions.
- Sepiolite increased the reversible thixotropy but also the irreversible one, which could reduce workability over time, but enhanced buildability.
- γ_{crit} can be defined as an effective parameter to characterise fresh CBM mechanical performance and the effects of components. In this study, γ_{crit} remained constant for each paste for 60 minutes.
- MC increased γ_{crit} producing more deformable pastes, and hence reducing buildability, while NC reduced γ_{crit} , increasing the elastic modulus and enhancing geometrical stability under load.
- This study has shown that cement fineness modifies the effect of admixtures. In addition, has been proven that a finer cement increased γ_{crit} due to the higher SP demand and affected shrinkage. Cement fineness is therefore an important factor to be considered in the design of 3DP cement-based mixtures, regarding both workability and buildability.
- NC remarkably increased A_{thix} and stiffening rate (ΔG) in pastes with a cement type 42.5 regarding 52.5 cement. This stiffening effect is related to the particle size and superficial interactions between cement grains and sepiolite.
- MC and NC combined in the proportion considered in this study resulted in negative interactions, related to the higher requirement of SP. However, it was observed that pastes with 42.5 cement and NC, increased thixotropy and stiffening rate compared to samples only with MC, which suggests that MC and NC combined in lower proportions could control 3D printing cement-based systems rheological performance.
- Early age parameters allowed to evaluate the influence of MC and NC on setting and the mechanical properties evolution.
- MC had a retardant effect on hydration kinetics and the development of mechanical properties, as they reduced the heat production and decreased the final UPV, while NC increased heat production and therefore promoted cement hydration.

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CRediT authorship contribution statement

Álvaro Márquez: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hugo Varela:** Writing – review & editing, Visualization, Methodology, Investigation, Conceptualization. **Gonzalo Barluenga:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

On my name and on behalf of my coauthors, I declare that there is not any conflict of interest with the contents of the manuscript we are submitting for possible publication in Construction and Building Materials Journal entitled “Rheology and early age evaluation of 3D printable cement-limestone filler pastes with Nanoclays and methylcellulose”.

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Data availability

Data will be made available on request.

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